

Exercise 5

What is a critical section? Describe it briefly: a critical section is a part of the process that accesses shared memory with other processes. This code can manipulate data of the shared memory causing data inconsistency.

What is a race condition? Please describe with your own words: A race condition is a result of critical code executing without proper synchronization on a shared memory and is a type of data inconsistency, where 2 processes work on the same data at the same time before one of them finishes causing the data to be corrupt. main problem is 2 different process or more manipulate shared data before either of them finishes.

What are the advantages of signaling and waiting compared to polling?: The main advantage is the CPU usage and efficiency while polling uses the CPU to keep checking whether a certain condition has been fulfilled or not. The sleep and signal sends the process to sleep waiting for a signal to continue, instead of waiting it uses the cpu to execute other processes until it gets the signal. This saves time, and avoids CPU overhead.

What is the difference between signaling and blocking?: Blocking = a state of a process where it cannot proceed because it's waiting for some event or resource. Signaling = an action performed by another process or the system that notifies the blocked process it can continue, changing its state from blocked → ready/running.

Study the code before we kill ourselves